



## OPERATOR GUIDE • QUIKBOLUS

# 3D Printing a Patient Bolus on the Bambu Lab P2S

A practical, step-by-step settings + supports guide for the operator who is printing a bolus from QuikBolus output for the first time. Tuned for TPU 95A on an enclosed P2S with the textured PEI plate.

Issued 2026-05-23 • Applies to Bambu Studio 1.9+ on Bambu Lab P2S

## 1. Pick the right material

Direct-print bolus material has to do three things at once: stay soft enough to conform to the patient, hold dimensional accuracy through dozens of fractions, and sit close enough to water-equivalent for the planner's dose calculation to stand up. **TPU 95A** is the right default for almost every patient bolus.

| MATERIAL                                                                  | USE IT FOR                                               | WHY / CAVEATS                                                                                                                                             |
|---------------------------------------------------------------------------|----------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>TPU 95A</b><br>( <i>Bambu, Polymaker PolyFlex, NinjaFlex Cheetah</i> ) | Default direct-print bolus                               | Skin-flexible, conforms well, density ~1.21 g/cc — close enough to water-equivalent for most build-up applications. Hygroscopic; dry before use.          |
| <b>TPU 85A</b>                                                            | Softer / more conformal bolus, very irregular anatomy    | Significantly harder to print — oozes, jams, prints slower. Don't start here. Earn it on simple geometries first.                                         |
| <b>PETG or PLA (<i>rigid</i>)</b>                                         | Mold for casting Flexibolus or similar pourable silicone | Not for direct patient contact. Smooth interior surface = clean cast. Print the mold rigid; cast the patient-facing bolus from water-equivalent silicone. |
| <b>ABS, PLA (<i>direct</i>)</b>                                           | —                                                        | <b>Avoid.</b> ABS warps and smells in an enclosed printer; PLA is too rigid for skin contact and not water-equivalent.                                    |

**AMS warning.** Bambu Lab themselves recommend running TPU from the **external spool holder**, not through the AMS. The AMS makes sharp bends that grip flexibles too hard and cause friction-jamming.

Disable AMS for the print and feed directly into the rear PTFE port.

## 2. Bambu Studio print profile — TPU 95A

Start from the built-in profile "0.20 mm Standard @BBL P2S 0.4 nozzle (TPU for AMS)" and override the values below. Save as a clinic-specific profile named *QuikBolus — TPU 95A — P2S* so the next operator gets the same settings.

| SETTING             | VALUE                          | WHY                                                                                                                                                                                                     |
|---------------------|--------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Layer height        | 0.20 mm                        | 0.16 for finer surface, 0.28 if speed matters. 0.20 is the comfort zone.                                                                                                                                |
| Wall loops          | 3                              | With solid infill, walls are partially redundant. 3 gives a clean rim without wasting print time.                                                                                                       |
| Top / bottom shells | 4                              | Watertight patient-contact surface. With 100% infill below, 4 is plenty.                                                                                                                                |
| Sparse infill       | 100%                           | <b>Clinical requirement.</b> The bolus must be dose-equivalent to water across its full thickness; sparse infill = air pockets = wrong CT number = wrong dose. See the "Why 100% infill" callout below. |
| Infill pattern      | Rectilinear<br>(or Concentric) | At 100% fill, pattern barely matters — pick a simple toolpath. Gyroid is overkill since there's no air to interlace through.                                                                            |
| Outer wall speed    | 25 mm/s                        | TPU loves slow outer walls — surface quality. Don't push past 30.                                                                                                                                       |
| Inner wall speed    | 40 mm/s                        |                                                                                                                                                                                                         |
| Infill speed        | 50 mm/s                        |                                                                                                                                                                                                         |
| First-layer speed   | 20 mm/s                        | TPU adhesion is fussy. Slow first layer = no curling.                                                                                                                                                   |
| Nozzle temperature  | 240 °C                         | 100% infill needs every bit of inter-layer adhesion it can get; run on the warmer end of the TPU range.                                                                                                 |
| Bed temperature     | 55 °C                          | Textured PEI plate. Bump to 60 °C if first layer lifts at corners.                                                                                                                                      |
| Retraction length   | 0.5–0.8 mm                     | TPU + long retraction = clog. Less is more.                                                                                                                                                             |
| Retraction speed    | 30 mm/s                        | Slow — don't snap the noodle back.                                                                                                                                                                      |
| Bridge speed        | 15 mm/s                        | TPU bridges OK if slow.                                                                                                                                                                                 |

| SETTING                   | VALUE                   | WHY                                                                                                                                                                 |
|---------------------------|-------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Fan speed (after layer 3) | 50 % infill / 80% walls | Solid fills need heat to bond layer-to-layer; aggressive cooling on infill = poor adhesion. Keep walls cool for definition. Layers 1–3 stay 0 % for plate adhesion. |
| Brim                      | 5 mm, outer-only        | TPU sometimes lifts at corners over a long print. Brim peels off cleanly post-print.                                                                                |
| Z-hop                     | OFF                     | Common source of TPU jams. Turn it off.                                                                                                                             |
| Pressure advance          | 0.045                   | P2S default for TPU. Tune $\pm 0.005$ if you see corner bulge (too high) or thin walls (too low).                                                                   |

**Why 100% infill is non-negotiable for a clinical bolus.** The dosimetrist's plan treats the bolus as homogeneous water-equivalent buildup. TPU 95A at 100% infill has density  $\sim 1.21$  g/cc, CT number  $\sim +75$  HU — close enough to water that the TPS dose calculation stands up. At 15% sparse infill, the bolus interior is mostly air (effective density  $\sim 0.18$  g/cc) and the volume-averaged density of a 5 mm bolus drops to  $\sim 0.6$  g/cc — half of water. Skin dose, depth-dose curve, electron range: all wrong. AAPM TG-176 calls out high infill ( $>90$  %) or solid as the only acceptable choice for dose-relevant 3D-printed bolus. The print-time cost is real but the dosimetric error of sparse infill is unacceptable.

**Run flow calibration for 100% infill specifically.** Bambu Studio → Calibrate → Flow Dynamics + Flow Rate. Over- extrusion is more obvious at 100% infill — you'll see bulging on top surfaces if flow is even a percent high. 15 minutes, done once per filament brand/spool.

### 3. Supports — the part everybody gets wrong

A patient bolus is anatomical, so the surface that touches the patient is rarely flat. Some overhangs are unavoidable. The trick is not "should I use supports" — the trick is **orient the part first**, then let the supports be minimal.

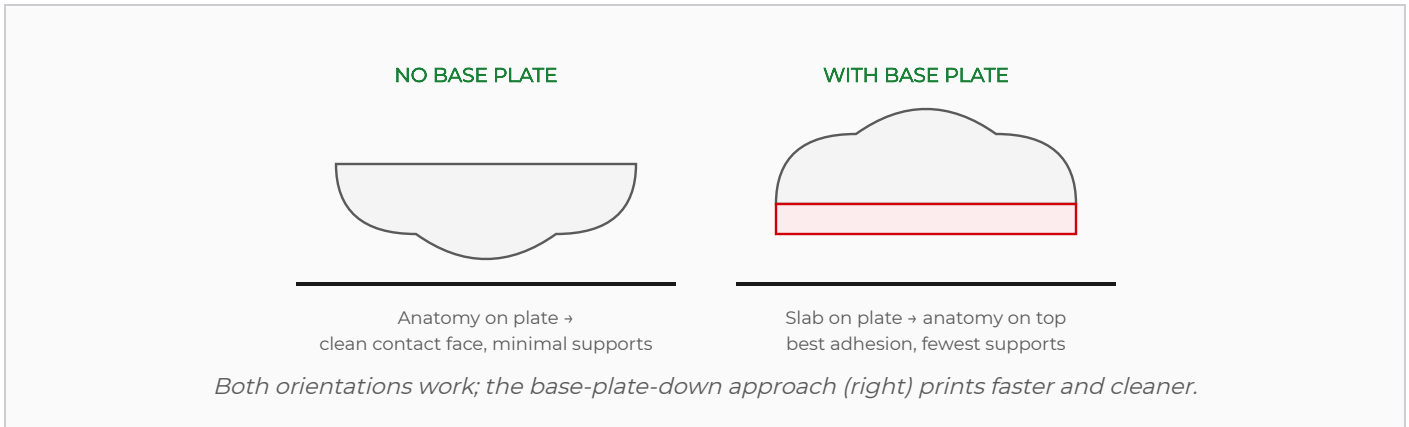
#### Step 1: Orient the part BEFORE generating supports

This is the single most important step and most people skip it. **What goes face-down on the build plate depends on whether you turned on QuikBolus's base-plate option:**

| QUIKBOLUS BASE PLATE   | BUILD-PLATE SIDE (DOWN)             | WHY                                                                                                                                                              |
|------------------------|-------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| OFF (no base plate)    | Anatomy (patient-contact) face DOWN | Build plate gives the cleanest finish — the patient-contact surface gets that finish.                                                                            |
| ON (base plate welded) | Base plate face DOWN                | Flat slab against the plate = perfect first-layer adhesion + the bolus body is fully supported by the slab below, so almost no tree supports needed. The concave |

| QUIKBOLUS BASE PLATE | BUILD-PLATE SIDE (DOWN) | WHY                                                                                                                                                                |
|----------------------|-------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                      |                         | patient surface ends up facing UP, which prints fine (concave-upward is a self-supporting shape, not an overhang). Same surface quality, less material, less time. |

**Practical default.** Turning the base plate ON in QuikBolus gives you the better print outcome regardless of whether you keep it on the final bolus — see section 4 for the keep-vs-trim discussion.



## Step 2: Support settings (TPU bolus)

| SETTING           | VALUE                  | WHY                                                                                                                    |
|-------------------|------------------------|------------------------------------------------------------------------------------------------------------------------|
| Support type      | Tree (organic)         | Less material, easier to remove than "normal". Marks TPU less.                                                         |
| Support style     | Snug                   | Tighter contour, less waste.                                                                                           |
| Top Z gap         | 0.25 mm                | TPU is the stickiest filament there is. Tighter than 0.25 = unremovable. 0.20 is too aggressive.                       |
| Interface layers  | 2                      | Smooth contact between support and part.                                                                               |
| Interface pattern | Rectilinear interlaced | Comes off in clean strips.                                                                                             |
| Threshold angle   | 30°                    | Overhangs steeper than 30° from vertical get supported.                                                                |
| Support filament  | Same TPU               | Single-material print. (Dual-material PLA-on-TPU would release cleaner, but AMS+TPU isn't recommended — not worth it.) |

## Step 3: When in doubt, paint supports instead of auto

For an anatomical bolus, auto-generated supports often add material where you don't actually need it. Use Bambu Studio's **Paint Supports** tool to mark only the regions that overhang. The patient-

contact surface (already on the plate, facing down) needs zero support. The outer dome typically needs minimal or no support — domes self-support up to ~45°.

**Rule of thumb.** A 5–10 mm thick bolus, oriented anatomy-side-down, usually needs supports **only at the perimeter rim** where it tapers to zero thickness, and possibly under deep concavities. The bulk of the part is self-supporting. If the auto-generator wants to fill the inside of your bolus with tree trunks, something's wrong — check orientation.

## 4. QuikBolus's base plate option — keep or trim?

QuikBolus can weld a flat slab onto the air-side face of the bolus. You then have a choice: **leave it on as part of the final bolus**, or **use it only as a printing aid and trim it off**. Both are valid clinical workflows; pick based on what your plan calls for.

### Method A — Keep it on (permanent bolus rigidity)

The base plate becomes part of the clinical bolus. It adds rigidity, gives a flat surface for handling and labeling, and lets you mount the bolus to a thermoplastic frame or indexed fixture.

- **Dosimetry note:** the base plate is in the beam path. It adds N mm of bolus material across the entire footprint — coordinate with the planner so the TPS calculation reflects the actual material.
- **Use when:** thin bolus (<5 mm) that needs stiffening; bolus that will be donned/doffed daily; you want to emboss patient ID on the flat side; the bolus mounts to a fixture.
- **Skip when:** the bolus is already self-supporting (>10 mm thick); electron treatment where the plan is to the millimeter; the planner can't re-plan to include the slab.

### Method B — Use as a print scaffold, then trim it off

Print with base plate ON because that's the best print orientation (see section 3). After the print, **trim the base plate off** with a hot knife, leaving only the anatomical bolus — which now exactly matches the dosimetrist's plan.

- **Advantage:** you get the print-process benefits (flat first layer, fully supported body, clean patient surface) AND the bolus you give the patient is just the anatomy — no dose discrepancy.
- **Disadvantage:** extra post-print step; trim accuracy matters (you want to cut flush with the original bolus contour, not at an arbitrary depth).

### How to trim cleanly

| TOOL                            | BEST FOR                      | TECHNIQUE                                                                                            |
|---------------------------------|-------------------------------|------------------------------------------------------------------------------------------------------|
| Hot knife                       | Any thickness (gold standard) | Pre-heat tip ~200 °C; slow, single pass; let the heat do the cut. TPU melts cleanly with no tearing. |
| Oscillating multi-tool (Dremel) | Thick base plates (>3 mm)     | Fine flush-cut blade; slow speed; light pressure. Wear safety glasses — TPU dust is fine and sticky. |
| Razor blade                     | Thin base plates (<1 mm)      | Fresh blade; long single stroke through the slab; the bolus geometry below acts as a guide.          |

| TOOL               | BEST FOR       | TECHNIQUE                                                                     |
|--------------------|----------------|-------------------------------------------------------------------------------|
| Sandpaper 220-grit | Edge finishing | After trimming, sand the cut edge smooth. Avoid sharp edges against the skin. |

**Pro trick:** before slicing, open the QuikBolus STL in Bambu Studio and use the *Cut* tool to add a thin score line (~0.5 mm groove) at the base-plate/bolus boundary. After printing, the groove guides your hot-knife stroke and you cut on exactly the right plane.

## Pick a method by scenario

| BOLUS SCENARIO                    | METHOD                 | BASE-PLATE THICKNESS           |
|-----------------------------------|------------------------|--------------------------------|
| < 5 mm bolus, head & neck patch   | A – keep               | 1 mm (just enough to handle)   |
| 5–8 mm bolus, frequent handling   | A – keep               | 1–2 mm                         |
| 5–8 mm bolus, plan is exact       | B – trim               | 2 mm (printing aid only)       |
| > 8 mm bolus, self-supporting     | B – trim or skip plate | 0–2 mm                         |
| HDR cap / chest-wall flash        | A – keep               | 2 mm (coordinate with planner) |
| Electron treatment, exact buildup | B – trim               | 2 mm (cut off post-print)      |

## 5. Bed prep — do these every time

- **Textured PEI plate** — never smooth PEI for TPU (sticks too well, you'll damage the contact surface removing it)
- Plate wiped with **isopropyl alcohol**; no glue stick, no hairspray, no Magigoo for TPU
- Filament **dried** if it's been open more than 2 weeks. Wet TPU = stringing + popping + voids. 60 °C for 6 hours in a filament dryer or food dehydrator
- AMS bypassed; filament loaded into the **rear PTFE port** directly from external spool holder
- **Flow calibration done** for this spool (Bambu Studio → Calibrate → Flow Dynamics + Flow Rate, ~15 min, once per spool)
- Bed leveled / auto-leveled within the last few prints; nozzle clean (no TPU bobble on the tip)

## 6. After the print — handling the finished bolus

1. **Remove from plate.** Flex the textured PEI plate gently — the bolus pops off. *Don't pry with a metal scraper*; you'll gouge the contact surface.
2. **Strip supports by hand.** Tree trunks pull off; interface layer peels. Stuck spots → flush cutters, never a knife.
3. **Inspect the patient-contact surface** under angled light. Slight plate texture transfer is fine and even desirable (small roughness helps skin adhesion). Hairline cracks or layer separation → reprint.
4. **Wash:** warm water + mild soap, rinse, isopropyl-wipe, air dry.
5. **Document** material lot, print profile name, print date, operator in the patient's chart. QuikBolus's audit record links to this.

**Do not autoclave TPU 95A.** It deforms above ~80 °C and the bolus loses its anatomical fit. Surface disinfection only (isopropyl, quaternary ammonium wipes). Single-patient use; discard after course completion or per your institutional policy.

## 7. When things go wrong

| SYMPTOM                                          | LIKELY CAUSE                         | FIX                                                                                     |
|--------------------------------------------------|--------------------------------------|-----------------------------------------------------------------------------------------|
| Print starts then jams a few millimeters in      | TPU bunching at extruder gear        | Lower retraction to 0.3 mm; reduce pressure advance; verify AMS bypass                  |
| Stringy gaps between perimeters                  | Too fast or too hot                  | Outer wall → 20 mm/s; nozzle -5 °C                                                      |
| Bolus warps off plate at hour 2                  | First-layer adhesion / bed too cold  | Brim 5 → 8 mm; bed +5 °C                                                                |
| Support fused permanently to part                | Z gap too small                      | 0.20 → 0.30 mm; consider lighter interface                                              |
| Patient-contact surface looks rough, layer-lined | Bridging over support gaps           | Re-orient: that surface goes ON the plate, not facing up                                |
| Print time "looks insane" (12+ hours)            | TPU is just slow at quality settings | This is normal. Don't chase speed; sacrifice surface quality. Plan around the duration. |
| Surface popping / micro-voids                    | Wet filament                         | Dry the spool 6 h @ 60 °C, reprint                                                      |
| Stringing between distant features               | Retraction set higher than TPU likes | 0.5 mm and slow (30 mm/s); some stringing on TPU is unavoidable                         |

## 8. Realistic print timing

TPU is slow. Plan accordingly — chasing speed costs you surface quality and adhesion, and there is no fast TPU.

| BOLUS GEOMETRY                              | APPROX PRINT TIME | FILAMENT USED |
|---------------------------------------------|-------------------|---------------|
| 5×7 cm × 5 mm thick (small head & neck)     | 8–12 hours        | 20–25 g       |
| 10×15 cm × 7 mm (breast / chest wall patch) | 20–28 hours       | 120–140 g     |
| 20×30 cm × 7 mm (full chest wall)           | 36–50 hours       | 480–560 g     |
| 20×30 cm × 10 mm (full chest wall, thick)   | 50–70 hours       | 680–800 g     |

Schedule strategy: head & neck patches start morning before clinic close; treat next morning. Breast/chest-wall patches start Friday evening for Monday treatment. **Full chest-wall prints (36–70 h) push the limits of direct printing** — consider switching to the mold + Flexibolus workflow (section 9) for those sizes, where total production (print rigid mold + pour + cure) is often faster than 50 h of solid TPU. Always print one spare per course in case of damage or fit issue.

**The print-time / dose-accuracy trade-off is real.** If you reduce infill to save time, you trade dose accuracy. There is no shortcut for a TPU bolus that preserves both. If print time genuinely cannot stretch to 100% solid, switch production methods (mold + Flexibolus cast) rather than reducing infill — same dose accuracy, different bottleneck.

## 9. Alternate workflow — print a mold, cast Flexibolus

For thicker bolus, build-up where water-equivalence matters most, or anatomy that wants the very softest conformance, an alternate workflow is to **3D-print a rigid mold** (PETG or PLA) and **cast the bolus from pourable Flexibolus** or similar tissue-equivalent silicone.

Trade-offs at a glance:

|                        | DIRECT-PRINT TPU 95A                | MOLD + FLEXIBOLUS CAST                                |
|------------------------|-------------------------------------|-------------------------------------------------------|
| Tissue-equivalent dose | Close (~1.21 g/cc)                  | Very close — Flexibolus is water-equivalent by design |
| Skin conformance       | Good                                | Excellent                                             |
| Production time        | Print only (single material change) | Print mold + cure silicone (hours added) + demold     |
| Reuse                  | Single-patient, single-course       | Mold lasts many casts; bolus is single-patient        |
| Operator time          | Low                                 | Higher — pour, degas, cure, demold                    |
| Surface finish         | Plate-side textured                 | Glass-smooth (the mold's inner surface transfers)     |



Use TPU 95A direct print for most cases. Reach for the mold-and-cast workflow when you need thick build-up, when water-equivalence matters more than usual (e.g., electron skin treatments, superficial photon fields), or when the anatomy is unusually irregular and you need a softer-than-TPU result.

## 10. Further reading & references

---

- **Bambu Studio 101 — Beginners Guide to Bambu Slicer Software** (YouTube): if Bambu Studio's interface is brand-new to you, watch this first. [youtube.com/watch?v=rZhy0J0mXBE](https://youtube.com/watch?v=rZhy0J0mXBE)
- **Bambu Lab TPU User Guide**: official guidance on AMS bypass, PTFE routing, and material handling. [wiki.bambulab.com](https://wiki.bambulab.com)
- **AAPM TG-176** (Olch et al.): dosimetric considerations for surface bolus, including discussion of bolus material density.
- **QuikBolus Users Guide**: end-to-end workflow from DICOM RT Structure Set to STL, including audit/traceability fields. [app.radiant-mpc.com/users-guides/quikbolus.pdf](https://app.radiant-mpc.com/users-guides/quikbolus.pdf)